User Service Installation on CentOS Stream 9 (Roboshop)

1. Install Node.js 20

The User service is developed using Node.js. Before deploying the application, install the supported Node.js runtime from the CentOS 9 AppStream repository.

```
dnf module disable nodejs -y
dnf module enable nodejs:20 -y
dnf install nodejs -y

node -v
npm -v
```

2. Create the Application User

For security reasons, applications should not run as the root user. Create a dedicated Linux user named **roboshop**. The command below creates it only if it does not already exist.

```
id roboshop &>/dev/null || useradd roboshop
```

3. Create the Application Directory

All application files will be stored under **/app**. The **-p** option prevents errors if the directory already exists.

```
mkdir -p /app
```

4. Download and Extract the Application

Download the User service artifact provided by the development team and extract it into the application directory.

```
curl -L -o /tmp/user.zip https://roboshop-builds.s3.amazonaws.com/user.zip

cd /app
unzip -o /tmp/user.zip
```

5. Install Application Dependencies

Every Node.js application defines its required libraries in **package.json**. Use **npm install** to download and install all dependencies.

```
cd /app
npm install
```

6. Configure the SystemD Service

Create a SystemD service so Linux can start, stop, restart and automatically enable the User service during boot. Replace the Redis and MongoDB placeholders with your server IP addresses.

```
vim /etc/systemd/system/user.service

[Unit]
Description=User Service

[Service]
User=roboshop
Environment=MONGO=true
Environment=REDIS_HOST=<REDIS-SERVER-IP>
Environment=MONGO_URL=mongodb://<MONGODB-SERVER-IP>:27017/users
ExecStart=/bin/node /app/server.js
SyslogIdentifier=user

[Install]
WantedBy=multi-user.target
```

7. Start the User Service

Reload SystemD so it detects the new service file, enable the service to start automatically after reboot, and start it immediately.

```
systemctl daemon-reload
systemctl enable user
systemctl start user
systemctl status user
```

8. Install MongoDB Shell

The User application stores data in MongoDB. To load the initial schema, install the MongoDB Shell (mongosh). First create the MongoDB repository and then install the package.

```
vim /etc/yum.repos.d/mongo.repo

[mongodb-org-8.0]
name=MongoDB Repository
baseurl=https://repo.mongodb.org/yum/redhat/9/mongodb-org/8.0/x86_64/
gpgcheck=0
enabled=1

dnf install mongodb-mongosh -y
```

9. Load the Database Schema

The schema files are shipped with the application source code. Load the schema into MongoDB using the following command.

```
mongosh --host <MONGODB-SERVER-IP> </app/schema/user.js
```

Command Summary

The following commands perform the complete installation from start to finish.

```
dnf module disable nodejs -y
dnf module enable nodejs:20 -y
dnf install nodejs -y

id roboshop &>/dev/null || useradd roboshop
mkdir -p /app

curl -L -o /tmp/user.zip https://roboshop-builds.s3.amazonaws.com/user.zip
cd /app
unzip -o /tmp/user.zip

npm install

systemctl daemon-reload
systemctl enable user
systemctl start user

dnf install mongodb-mongosh -y
mongosh --host <MONGODB-SERVER-IP> </app/schema/user.js
```